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DNN-based cross component prediction

Abstract

Systems and methods for deep neural network (DNN)-based cross component prediction are provided. A method includes inputting a reconstructed luma block of an image or video into a DNN; and predicting, by the DNN, a reconstructed chroma block of the image or video based on the reconstructed luma block that is input. Luma and chroma reference information and side information may also be input into the DNN to predict the reconstructed chroma block. The various inputs may also be generated using processes such as downsampling and transformation.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
2017/0230656	12/2016	Leontaris et al.	N/A	N/A
2020/0186809	12/2019	Mukherjee et al.	N/A	N/A
2020/0304833	12/2019	Budagavi	N/A	N/A
2023/0016377	12/2022	Zhu et al.	N/A	N/A
2023/0336736	12/2022	Gao	N/A	H04N 19/172

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
110602491	12/2018	CN	N/A
2020/264457	12/2019	WO	N/A
2021/069688	12/2020	WO	N/A

OTHER PUBLICATIONS

Extended European Search Report issued Jul. 24, 2023 in European Application No. 22790194.9. cited by applicant

Liqiang Wang, et al., “AHG11:neural network based cross-component prediction model”, Joint Video Experts Team (JVET) of ITU-T SG16 WP 3 and ISO/IEC JTC 1/SC 29, JVET-W0111-v2, Jul. 7-16, 2021, 23rd Meeting, by teleconference (5 pages). cited by applicant

Young-Yoon Lee, et al., “AHG11: Cross-component prediction based on a neural network model”, Joint Video Experts Team (JVET) of ITU-T SG 16 WP 3 and ISO/IEC JTC 1/SC 29, JVET-X0130-v1, Oct. 11-15, 2021, 24th Meeting, by teleconference (5 pages). cited by applicant

Marc Gorriz Blanch, et al., “Chroma Intra Prediction With Attention-Based CNN Architectures”, IEEE, ICIP, 2020, pp. 783-787 (5 pages). cited by applicant

Yue Li, et al., “A Hybrid Neural Network for Chorma Intra Prediction”, IEEE, ICIP, 2018, pp. 1797-1801 (5 pages). cited by applicant

International Search Report dated Oct. 19, 2022 in Application No. PCT/US22/31506. cited by applicant

Written Opinion of the International Searching Authority dated Oct. 19, 2022 in Application No. PCT/US22/31506. cited by applicant

Marc Gorriz Blanch et al., “Attention-Based Neural Networks for Chroma Intra Prediction in Video Coding”, Journal of Selected Topics in Signal Processing, Oct. 2020, pp. 1-12 (12 pages total). cited by applicant
Linwei Zhu, et al., “Deep Learning-Based Chroma Prediction for Intra Versatile Video Coding”, IEEE Transactions on Circuits and Systems for Video Technology, Aug. 2021, vol. 31, No. 8, pp. 3168-3181 (14 pages). cited by applicant
K. Zhang, J. Chen, L. Zhang, X. Li and M. Karczewicz, “Enhanced Cross-Component Linear Model for Chroma Intra-Prediction in Video Coding,” in IEEE Transactions on Image Processing, vol. 27, No. 8, pp. 3983-3997, Aug. 2018, doi: 10.1109/TIP .2018.2830640. (Year: 2018). cited by applicant

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION (1) This application is a continuation of U.S. application Ser. No. 17/749,730, filed May 20, 2022, which claims priority from U.S. Provisional Application No. 63/210,741, filed on Jun. 15, 2021, the disclosure of which is incorporated herein by reference in its entirety.

FIELD
(1) Embodiments of the present disclosure are directed to methods and systems of DNN-based cross component prediction.
BACKGROUND
(2) Traditional video coding standards, such as the H.264/Advanced Video Coding (H.264/AVC), High-Efficiency Video Coding (HEVC), and Versatile Video Coding (VVC) are designed on a similar (recursive) block-based hybrid prediction/transform framework where individual coding tools like intra/inter prediction, integer transforms, and context-adaptive entropy coding, are intensively handcrafted to optimize the overall efficiency. Basically, the spatiotemporal pixel neighborhoods are leveraged for predictive signal construction, to obtain corresponding residuals for subsequent transform, quantization, and entropy coding. On the other hand, the nature of Deep Neural Networks (DNN) is to extract different levels of spatiotemporal stimuli by analyzing spatiotemporal information from the receptive field of neighboring pixels. The capability of exploring highly nonlinearity and nonlocal spatiotemporal correlations provide promising opportunity for largely improved compression quality.
(3) One purpose of video coding and decoding can be the reduction of redundancy in the input video signal, through compression. Compression can help reducing aforementioned bandwidth or storage space requirements, in some cases by two orders of magnitude or more. Both lossless and lossy compression, as well as a combination thereof can be employed. Lossless compression refers to techniques where an exact copy of the original signal can be reconstructed from the compressed original signal. When using lossy compression, the reconstructed signal may not be identical to the original signal, but the distortion between original and reconstructed signal is small enough to make the reconstructed signal useful for the intended application. In the case of video, lossy compression is widely employed. The amount of distortion tolerated depends on the application; for example, users of certain consumer streaming applications may tolerate higher distortion than users of television contribution applications. The compression ratio achievable can reflect that: higher allowable/tolerable distortion can yield higher compression ratios.

SUMMARY

(4) Leveraging information from different components and other side information, a traditional encoder can predict other components to achieve better compression performance. However, cross component linear prediction mode in intra-prediction cannot work well compared with a DNN-based method. The nature of DNN is to extract different high level of stimuli and the capability of exploring highly nonlinearity and nonlocal correlations provides promising opportunity for high compression quality. Embodiments of the present disclosure use a DNN-based model to handle an arbitrary shape of luma component, reference components, and side information to predict a reconstructed chroma component to achieve better compression performance.

(5) Embodiments of the present disclosure provide a Cross Component Prediction (CCP) model as a new mode in intra prediction by using a Deep Neural Network (DNN). The model uses the information provided by the encoder, such as the luma component, quantization parameter (QP) value, block depth, etc., to predict chroma component to achieve better compression performance. Previous NN-based intra prediction approaches are either aimed only at the prediction of luma component or generate the prediction for all three channels, disregarding the correlation between chroma component and other additional information.

(6) According to embodiments, a method performed by at least one processor is provided. The method includes: obtaining a reconstructed luma block of an image or video; inputting the reconstructed luma block into a DNN; obtaining reference components and side information associated with the reconstructed luma block; inputting the reference components and the side information into the DNN; and predicting, by the DNN, a reconstructed chroma block of the image or video based on the reconstructed luma block, the reference components, and the side information.

(7) According to embodiments, a system is provided. The system includes: at least one memory configured to store computer program code; and at least one processor configured to access the computer program code and operate as instructed by the computer program code. The compute program code includes: inputting code configured to cause the at least one processor to input a reconstructed luma block of an image or video, reference components, and side information associated with the reconstructed luma block into a deep neural network (DNN) that is implemented by the at least one processor; and predicting code configured to cause the at least one processor to predict, by the DNN, a reconstructed chroma block of the image or video based on the reconstructed luma block, the reference components, and the side information that are input.

(8) According to embodiments, a non-transitory computer-readable medium storing computer code is provided. The computer code is configured to, when executed by at least one processor, cause the at least one processor to: implement a DNN; input a reconstructed luma block of an image or video, reference components, and side information associated with the reconstructed luma block into the DNN; and predict, by the DNN, a reconstructed chroma block of the image or video based on the reconstructed luma block, the reference components, and the side information that are input.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Further features, the nature, and various advantages of the disclosed subject matter will be more apparent from the following detailed description and the accompanying drawings in which:

(2) FIG. **1** is a schematic illustration of a simplified block diagram of a communication system in accordance with an embodiment.

(3) FIG. **2** is a schematic illustration of a simplified block diagram of a communication system in accordance with an embodiment.

(4) FIG. **3** is a schematic illustration of a simplified block diagram of a decoder in accordance with

an embodiment.
(5) FIG. **4** is a schematic illustration of a simplified block diagram of an encoder in accordance with an embodiment.
(6) FIG. **5** is a schematic illustration of a simplified block diagram of a process of input generation in accordance with an embodiment.
(7) FIG. **6** is a schematic illustration of a simplified block diagram of a process of cross-component prediction in accordance with an embodiment.
(8) FIG. **7** is a block diagram of computer code according to embodiments.
(9) FIG. **8** is a diagram of a computer system suitable for implementing embodiments of the present disclosure.

DETAILED DESCRIPTION

(10) FIG. **1** illustrates a simplified block diagram of a communication system **100** according to an embodiment of the present disclosure. The communication system **100** may include at least two terminals **110**, **120** interconnected via a network **150**. For unidirectional transmission of data, a first terminal **110** may code video data at a local location for transmission to the other terminal **120** via the network **150**. The second terminal **120** may receive the coded video data of the other terminal from the network **150**, decode the coded data and display the recovered video data. Unidirectional data transmission may be common in media serving applications and the like.
(11) FIG. **1** illustrates a second pair of terminals **130**, **140** provided to support bidirectional transmission of coded video that may occur, for example, during videoconferencing. For bidirectional transmission of data, each terminal **130**, **140** may code video data captured at a local location for transmission to the other terminal via the network **150**. Each terminal **130**, **140** also may receive the coded video data transmitted by the other terminal, may decode the coded data, and may display the recovered video data at a local display device.
(12) In FIG. **1**, the terminals **110-140** may be illustrated as servers, personal computers, and smart phones, and/or any other type of terminal. For example, the terminals **110-140** may be laptop computers, tablet computers, media players and/or dedicated video conferencing equipment. The network **150** represents any number of networks that convey coded video data among the terminals **110-140**, including for example wireline and/or wireless communication networks. The communication network **150** may exchange data in circuit-switched and/or packet-switched channels. Representative networks include telecommunications networks, local area networks, wide area networks, and/or the Internet. For the purposes of the present discussion, the architecture and topology of the network **150** may be immaterial to the operation of the present disclosure unless explained herein below.
(13) FIG. **2** illustrates, as an example for an application for the disclosed subject matter, the placement of a video encoder and decoder in a streaming environment. The disclosed subject matter can be equally applicable to other video enabled applications, including, for example, video conferencing, digital TV, storing of compressed video on digital media including CD, DVD, memory stick and the like, and so on.
(14) As illustrated in FIG. **2**, a streaming system **200** may include a capture subsystem **213** that can include a video source **201** and an encoder **203**. The video source **201** may be, for example, a digital camera, and may be configured to create an uncompressed video sample stream **202**. The uncompressed video sample stream **202** may provide a high data volume when compared to encoded video bitstreams, and can be processed by the encoder **203** coupled to the video source **201**. The encoder **203** can include hardware, software, or a combination thereof to enable or implement aspects of the disclosed subject matter as described in more detail below. The encoded video bitstream **204** may include a lower data volume when compared to the sample stream, and can be stored on a streaming server **205** for future use. One or more streaming clients **206** can access the streaming server **205** to retrieve video bitstreams **209** that may be copies of the encoded video bitstream **204**.

(15) In embodiments, the streaming server **205** may also function as a Media-Aware Network Element (MANE). For example, the streaming server **205** may be configured to prune the encoded video bitstream **204** for tailoring potentially different bitstreams to one or more of the streaming clients **206**. In embodiments, a MANE may be separately provided from the streaming server **205** in the streaming system **200**.
(16) The streaming clients **206** can include a video decoder **210** and a display **212**. The video decoder **210** can, for example, decode video bitstream **209**, which is an incoming copy of the encoded video bitstream **204**, and create an outgoing video sample stream **211** that can be rendered on the display **212** or another rendering device (not depicted). In some streaming systems, the video bitstreams **204**, **209** can be encoded according to certain video coding/compression standards. Examples of such standards include, but are not limited to, ITU-T Recommendation H.265. Under development is a video coding standard informally known as Versatile Video Coding (VVC). Embodiments of the disclosure may be used in the context of VVC.
(17) FIG. **3** illustrates an example functional block diagram of a video decoder **210** that is attached to a display **212** according to an embodiment of the present disclosure.
(18) The video decoder **210** may include a channel **312**, receiver **310**, a buffer memory **315**, an entropy decoder/parser **320**, a scaler/inverse transform unit **351**, an intra picture prediction unit **352**, a Motion Compensation Prediction unit **353**, an aggregator **355**, a loop filter unit **356**, reference picture memory **357**, and current picture memory. In at least one embodiment, the video decoder **210** may include an integrated circuit, a series of integrated circuits, and/or other electronic circuitry. The video decoder **210** may also be partially or entirely embodied in software running on one or more CPUs with associated memories.
(19) In this embodiment, and other embodiments, the receiver **310** may receive one or more coded video sequences to be decoded by the decoder **210** one coded video sequence at a time, where the decoding of each coded video sequence is independent from other coded video sequences. The coded video sequence may be received from the channel **312**, which may be a hardware/software link to a storage device which stores the encoded video data. The receiver **310** may receive the encoded video data with other data, for example, coded audio data and/or ancillary data streams, that may be forwarded to their respective using entities (not depicted). The receiver **310** may separate the coded video sequence from the other data. To combat network jitter, the buffer memory **315** may be coupled in between the receiver **310** and the entropy decoder/parser **320** ("parser" henceforth). When the receiver **310** is receiving data from a store/forward device of sufficient bandwidth and controllability, or from an isosynchronous network, the buffer memory **315** may not be used, or can be small. For use on best effort packet networks such as the Internet, the buffer memory **315** may be required, can be comparatively large, and can be of adaptive size.
(20) The video decoder **210** may include a parser **320** to reconstruct symbols **321** from the entropy coded video sequence. Categories of those symbols include, for example, information used to manage operation of the decoder **210**, and potentially information to control a rendering device such as a display **212** that may be coupled to a decoder as illustrated in FIG. **2**. The control information for the rendering device(s) may be in the form of, for example, Supplementary Enhancement Information (SEI) messages or Video Usability Information (VUI) parameter set fragments (not depicted). The parser **320** may parse/entropy-decode the coded video sequence received. The coding of the coded video sequence can be in accordance with a video coding technology or standard, and can follow principles well known to a person skilled in the art, including variable length coding, Huffman coding, arithmetic coding with or without context sensitivity, and so forth. The parser **320** may extract from the coded video sequence, a set of subgroup parameters for at least one of the subgroups of pixels in the video decoder, based upon at least one parameters corresponding to the group. Subgroups can include Groups of Pictures (GOPs), pictures, tiles, slices, macroblocks, Coding Units (CUs), blocks, Transform Units (TUs), Prediction Units (PUs) and so forth. The parser **320** may also extract from the coded video

sequence information such as transform coefficients, quantizer parameter values, motion vectors, and so forth.

(21) The parser **320** may perform entropy decoding/parsing operation on the video sequence received from the buffer memory **315**, so to create symbols **321**.

(22) Reconstruction of the symbols **321** can involve multiple different units depending on the type of the coded video picture or parts thereof (such as: inter and intra picture, inter and intra block), and other factors. Which units are involved, and how they are involved, can be controlled by the subgroup control information that was parsed from the coded video sequence by the parser **320**. The flow of such subgroup control information between the parser **320** and the multiple units below is not depicted for clarity.

(23) Beyond the functional blocks already mentioned, decoder **210** can be conceptually subdivided into a number of functional units as described below. In a practical implementation operating under commercial constraints, many of these units interact closely with each other and can, at least partly, be integrated into each other. However, for the purpose of describing the disclosed subject matter, the conceptual subdivision into the functional units below is appropriate.

(24) One unit may be the scaler/inverse transform unit **351**. The scaler/inverse transform unit **351** may receive quantized transform coefficient as well as control information, including which transform to use, block size, quantization factor, quantization scaling matrices, etc. as symbol(s) **321** from the parser **320**. The scaler/inverse transform unit **351** can output blocks including sample values that can be input into the aggregator **355**.

(25) In some cases, the output samples of the scaler/inverse transform unit **351** can pertain to an intra coded block; that is: a block that is not using predictive information from previously reconstructed pictures, but can use predictive information from previously reconstructed parts of the current picture. Such predictive information can be provided by an intra picture prediction unit **352**. In some cases, the intra picture prediction unit **352** generates a block of the same size and shape of the block under reconstruction, using surrounding already reconstructed information fetched from the current (partly reconstructed) picture from the current picture memory **358**. The aggregator **355**, in some cases, adds, on a per sample basis, the prediction information the intra picture prediction unit **352** has generated to the output sample information as provided by the scaler/inverse transform unit **351**.

(26) In other cases, the output samples of the scaler/inverse transform unit **351** can pertain to an inter coded, and potentially motion compensated block. In such a case, a Motion Compensation Prediction unit **353** can access reference picture memory **357** to fetch samples used for prediction. After motion compensating the fetched samples in accordance with the symbols **321** pertaining to the block, these samples can be added by the aggregator **355** to the output of the scaler/inverse transform unit **351** (in this case called the residual samples or residual signal) so to generate output sample information. The addresses within the reference picture memory **357**, from which the Motion Compensation Prediction unit **353** fetches prediction samples, can be controlled by motion vectors. The motion vectors may be available to the Motion Compensation Prediction unit **353** in the form of symbols **321** that can have, for example, X, Y, and reference picture components. Motion compensation also can include interpolation of sample values as fetched from the reference picture memory **357** when sub-sample exact motion vectors are in use, motion vector prediction mechanisms, and so forth.

(27) The output samples of the aggregator **355** can be subject to various loop filtering techniques in the loop filter unit **356**. Video compression technologies can include in-loop filter technologies that are controlled by parameters included in the coded video bitstream and made available to the loop filter unit **356** as symbols **321** from the parser **320**, but can also be responsive to meta-information obtained during the decoding of previous (in decoding order) parts of the coded picture or coded video sequence, as well as responsive to previously reconstructed and loop-filtered sample values.

(28) The output of the loop filter unit **356** can be a sample stream that can be output to a render

device such as a display **212**, as well as stored in the reference picture memory **357** for use in future inter-picture prediction.
(29) Certain coded pictures, once fully reconstructed, can be used as reference pictures for future prediction. Once a coded picture is fully reconstructed and the coded picture has been identified as a reference picture (by, for example, parser **320**), the current reference picture can become part of the reference picture memory **357**, and a fresh current picture memory can be reallocated before commencing the reconstruction of the following coded picture.
(30) The video decoder **210** may perform decoding operations according to a predetermined video compression technology that may be documented in a standard, such as ITU-T Rec. H.265. The coded video sequence may conform to a syntax specified by the video compression technology or standard being used, in the sense that it adheres to the syntax of the video compression technology or standard, as specified in the video compression technology document or standard and specifically in the profiles document therein. Also, for compliance with some video compression technologies or standards, the complexity of the coded video sequence may be within bounds as defined by the level of the video compression technology or standard. In some cases, levels restrict the maximum picture size, maximum frame rate, maximum reconstruction sample rate (measured in, for example megasamples per second), maximum reference picture size, and so on. Limits set by levels can, in some cases, be further restricted through Hypothetical Reference Decoder (HRD) specifications and metadata for HRD buffer management signaled in the coded video sequence.
(31) In an embodiment, the receiver **310** may receive additional (redundant) data with the encoded video. The additional data may be included as part of the coded video sequence(s). The additional data may be used by the video decoder **210** to properly decode the data and/or to more accurately reconstruct the original video data. Additional data can be in the form of, for example, temporal, spatial, or SNR enhancement layers, redundant slices, redundant pictures, forward error correction codes, and so on.
(32) FIG. **4** illustrates an example functional block diagram of a video encoder **203** associated with a video source **201** according to an embodiment of the present disclosure.
(33) The video encoder **203** may include, for example, an encoder that is a source coder **430**, a coding engine **432**, a (local) decoder **433**, a reference picture memory **434**, a predictor **435**, a transmitter **440**, an entropy coder **445**, a controller **450**, and a channel **460**.
(34) The encoder **203** may receive video samples from a video source **201** (that is not part of the encoder) that may capture video image(s) to be coded by the encoder **203**.
(35) The video source **201** may provide the source video sequence to be coded by the encoder **203** in the form of a digital video sample stream that can be of any suitable bit depth (for example: 8 bit, 10 bit, 12 bit, . . .), any colorspace (for example, BT.601 Y CrCB, RGB, . . .) and any suitable sampling structure (for example Y CrCb 4:2:0, Y CrCb 4:4:4). In a media serving system, the video source **201** may be a storage device storing previously prepared video. In a videoconferencing system, the video source **201** may be a camera that captures local image information as a video sequence. Video data may be provided as a plurality of individual pictures that impart motion when viewed in sequence. The pictures themselves may be organized as a spatial array of pixels, wherein each pixel can include one or more sample depending on the sampling structure, color space, etc. in use. A person skilled in the art can readily understand the relationship between pixels and samples. The description below focuses on samples.
(36) According to an embodiment, the encoder **203** may code and compress the pictures of the source video sequence into a coded video sequence **443** in real time or under any other time constraints as required by the application. Enforcing appropriate coding speed is one function of controller **450**. The controller **450** may also control other functional units as described below and may be functionally coupled to these units. The coupling is not depicted for clarity. Parameters set by the controller **450** can include rate control related parameters (picture skip, quantizer, lambda value of rate-distortion optimization techniques, . . .), picture size, group of pictures (GOP) layout,

maximum motion vector search range, and so forth. A person skilled in the art can readily identify other functions of controller **450** as they may pertain to video encoder **203** optimized for a certain system design.

(37) Some video encoders operate in what a person skilled in the are readily recognizes as a "coding loop". As an oversimplified description, a coding loop can consist of the encoding part of the source coder **430** (responsible for creating symbols based on an input picture to be coded, and a reference picture(s)), and the (local) decoder **433** embedded in the encoder **203** that reconstructs the symbols to create the sample data that a (remote) decoder also would create when a compression between symbols and coded video bitstream is lossless in certain video compression technologies. That reconstructed sample stream may be input to the reference picture memory **434**. As the decoding of a symbol stream leads to bit-exact results independent of decoder location (local or remote), the reference picture memory content is also bit exact between a local encoder and a remote encoder. In other words, the prediction part of an encoder "sees" as reference picture samples exactly the same sample values as a decoder would "see" when using prediction during decoding. This fundamental principle of reference picture synchronicity (and resulting drift, if synchronicity cannot be maintained, for example because of channel errors) is known to a person skilled in the art.

(38) The operation of the "local" decoder **433** can be the same as of a "remote" decoder **210**, which has already been described in detail above in conjunction with FIG. **3**. However, as symbols are available and en/decoding of symbols to a coded video sequence by the entropy coder **445** and the parser **320** can be lossless, the entropy decoding parts of decoder **210**, including channel **312**, receiver **310**, buffer memory **315**, and parser **320** may not be fully implemented in the local decoder **433**.

(39) An observation that can be made at this point is that any decoder technology, except the parsing/entropy decoding that is present in a decoder, may need to be present, in substantially identical functional form in a corresponding encoder. For this reason, the disclosed subject matter focuses on decoder operation. The description of encoder technologies can be abbreviated as they may be the inverse of the comprehensively described decoder technologies. Only in certain areas a more detail description is required and provided below.

(40) As part of its operation, the source coder **430** may perform motion compensated predictive coding, which codes an input frame predictively with reference to one or more previously-coded frames from the video sequence that were designated as "reference frames." In this manner, the coding engine **432** codes differences between pixel blocks of an input frame and pixel blocks of reference frame(s) that may be selected as prediction reference(s) to the input frame.

(41) The local decoder **433** may decode coded video data of frames that may be designated as reference frames, based on symbols created by the source coder **430**. Operations of the coding engine **432** may advantageously be lossy processes. When the coded video data may be decoded at a video decoder (not shown in FIG. **4**), the reconstructed video sequence typically may be a replica of the source video sequence with some errors. The local decoder **433** replicates decoding processes that may be performed by the video decoder on reference frames and may cause reconstructed reference frames to be stored in the reference picture memory **434**. In this manner, the encoder **203** may store copies of reconstructed reference frames locally that have common content as the reconstructed reference frames that will be obtained by a far-end video decoder (absent transmission errors).

(42) The predictor **435** may perform prediction searches for the coding engine **432**. That is, for a new frame to be coded, the predictor **435** may search the reference picture memory **434** for sample data (as candidate reference pixel blocks) or certain metadata such as reference picture motion vectors, block shapes, and so on, that may serve as an appropriate prediction reference for the new pictures. The predictor **435** may operate on a sample block-by-pixel block basis to find appropriate prediction references. In some cases, as determined by search results obtained by the predictor **435**,

an input picture may have prediction references drawn from multiple reference pictures stored in the reference picture memory **434**.
(43) The controller **450** may manage coding operations of the source coder **430**, including, for example, setting of parameters and subgroup parameters used for encoding the video data.
(44) Output of all aforementioned functional units may be subjected to entropy coding in the entropy coder **445**. The entropy coder translates the symbols as generated by the various functional units into a coded video sequence, by loss-less compressing the symbols according to technologies known to a person skilled in the art as, for example Huffman coding, variable length coding, arithmetic coding, and so forth.
(45) The transmitter **440** may buffer the coded video sequence(s) as created by the entropy coder **445** to prepare it for transmission via a communication channel **460**, which may be a hardware/software link to a storage device which would store the encoded video data. The transmitter **440** may merge coded video data from the source coder **430** with other data to be transmitted, for example, coded audio data and/or ancillary data streams (sources not shown).
(46) The controller **450** may manage operation of the encoder **203**. During coding, the controller **450** may assign to each coded picture a certain coded picture type, which may affect the coding techniques that may be applied to the respective picture. For example, pictures often may be assigned as an Intra Picture (I picture), a Predictive Picture (P picture), or a Bi-directionally Predictive Picture (B Picture).
(47) An Intra Picture (I picture) may be one that may be coded and decoded without using any other frame in the sequence as a source of prediction. Some video codecs allow for different types of Intra pictures, including, for example Independent Decoder Refresh (IDR) Pictures. A person skilled in the art is aware of those variants of I pictures and their respective applications and features.
(48) A Predictive picture (P picture) may be one that may be coded and decoded using intra prediction or inter prediction using at most one motion vector and reference index to predict the sample values of each block.
(49) A Bi-directionally Predictive Picture (B Picture) may be one that may be coded and decoded using intra prediction or inter prediction using at most two motion vectors and reference indices to predict the sample values of each block. Similarly, multiple-predictive pictures can use more than two reference pictures and associated metadata for the reconstruction of a single block.
(50) Source pictures commonly may be subdivided spatially into a plurality of sample blocks (for example, blocks of 4×4, 8×8, 4×8, or 16×16 samples each) and coded on a block-by-block basis. Blocks may be coded predictively with reference to other (already coded) blocks as determined by the coding assignment applied to the blocks' respective pictures. For example, blocks of I pictures may be coded non-predictively or they may be coded predictively with reference to already coded blocks of the same picture (spatial prediction or intra prediction). Pixel blocks of P pictures may be coded non-predictively, via spatial prediction or via temporal prediction with reference to one previously coded reference pictures. Blocks of B pictures may be coded non-predictively, via spatial prediction or via temporal prediction with reference to one or two previously coded reference pictures.
(51) The video encoder **203** may perform coding operations according to a predetermined video coding technology or standard, such as ITU-T Rec. H.265. In its operation, the video encoder **203** may perform various compression operations, including predictive coding operations that exploit temporal and spatial redundancies in the input video sequence. The coded video data, therefore, may conform to a syntax specified by the video coding technology or standard being used.
(52) In an embodiment, the transmitter **440** may transmit additional data with the encoded video. The source coder **430** may include such data as part of the coded video sequence. Additional data may comprise temporal/spatial/SNR enhancement layers, other forms of redundant data such as redundant pictures and slices, Supplementary Enhancement Information (SEI) messages, Visual

Usability Information (VUI) parameter set fragments, and so on.
(53) Embodiments of the present disclosure provide DNN-based cross component prediction. Examples embodiments are described below with reference to FIGS. **5-6**.
(54) A video compression framework, according to embodiments of the present disclosure, is described as follows. Assume an input video includes a plurality of image frames equal to a total number of frames in a video. The frames are partitioned into spatial blocks, each block can be partitioned into smaller blocks iteratively. The block contains both a luma component **510***y* and a chroma component **520** that includes chroma channels **520***u* and **520***t*. During an intra-prediction process, the luma component **510***y* can be predicted first, and then the two chroma channels **520***u* and **520***t* can be predicted later. The prediction of both chroma channels **520***u* and **520***t* can be performed jointly or separately.
(55) In an embodiment of the present disclosure, the reconstructed chroma component **520** is generated by DNN-based models in both an encoder and a decoder, or only in the decoder. The two chroma channels **520***u* and **520***t* may be generated together with a single network, or separately with different networks. For each chroma channel, the chroma channel may be generated using a different network based on the block size. One or more processes including signal-processing, spatial or temporal filtering, scaling, weighted averaging, up-/down-sampling, pooling, recursive processing with memory, linear system processing, non-linear system processing, neural-network processing, deep-learning based processing, AI-processing, pre-trained network processing, machine-learning based processing, or their combinations can be used as modules in embodiments of the present disclosure in the DNN-based cross component prediction. For processing the reconstructed chroma component **520**, one reconstructed chroma channel (e.g., one from among the chroma channels **520***u* and **520***t*) can be used to generate the other reconstructed chroma channel (e.g., the other from among the chroma channels **520***u* and **520***t*).
(56) According to embodiments of the present disclosure, a DNN-based Cross Component Prediction model may be provided that enhances the compression performance of reconstructed chroma channels **520***u* and **520***t* of a block, based on a reconstructed luma component **510***y* of the block, reference components, and other side information provided by the encoder. According to embodiments, 4:2:0 may be used for subsampling the chroma channels **520***u* and **520***t*. Therefore, the chroma channels **520***u* and **520***t* may have a lower resolution than the luma component **510***y*.
(57) With reference to FIG. **5**, a process **500** is described below. The process **500** includes a workflow of generating input samples **580** for training and/or prediction in a general hybrid video coding system according to embodiments of the present disclosure.
(58) The reconstructed luma component **510***y* may be a luma block that is a 2N×2M block, wherein 2N is the width of the luma block and 2M is the height of the luma block. According to embodiments, a first luma reference **512***y* that is a 2N×2K block, and a second luma reference **514***y* that is 2K×2M block may also be provided, wherein 2K stands for the number of rows or columns in the luma references. To make the luma size the same as a predicted output size, a downsampling process **591** is applied for the luma component **510***y*, the first luma reference **512***y*, and the second luma reference **514***y*. The downsampling process **530** can be a traditional method such as bicubic and bilinear, or it can be NN-based downsampling method. After the downsampling, the luma component **510***y* may become a downsampled luma component **530***y* having a block size of N×M, the first luma reference **512***y* may become a downsampled first luma reference **532***y* having a block size of N×K, and the second luma reference **514***y* may become a downsampled second luma reference **534***y* having a block size of K×M. The downsampled first luma reference **532***y* and the downsampled second luma reference **534***y* may be transformed (at step **592**) to become a first transformed luma reference **552***y* and a second transformed luma reference **554***y*, respectively, that match the size of the downsampled luma component **530***y* (also referred to as a luma block), and the first transformed luma reference **552***y*, the second transformed luma reference **554***y*, and the downsampled luma component **530***y* may be concatenated (at step **592**) together. For example, the

transformation can be performed by duplicating the value of the downsampled first luma reference **532***y* and the downsampled second luma reference **534***y* several times until the size thereof are the same as an output block size (e.g., the size of the downsampled luma component **530***y*).
(59) To predict the chroma component **520**, adjacent references (e.g., a first chroma reference **522** and a second chroma reference **524**) of the chroma component **520** can also be added as an optional reference for generating a better chroma component. With reference to FIG. **5**, the chroma component **520** may be a block having a size N×M, which is the reconstructed chroma block that may be generated/predicted in embodiments of the present disclosure. The chroma component **520** has two chroma channels **520***u* and **520***t*, and both channels **520***u* and **520***t* may be used jointly. The first chroma reference **522** and the second chroma reference **524** may be obtained (at step **593**), which may have a block size of N×K and K×M, respectively. According to embodiments, the first chroma reference **522** and the second chroma reference **524** may each be obtained twice to correspond to the two chroma channels **520***u* and **520***t*. The first chroma reference **522** and the second chroma reference **524** may be transformed (at step **594**) to a first transformed chroma reference **542** and a second transformed chroma reference **544**, respectively, that match the size of N×M. All image-based information (e.g., the downsampled luma component **530***y*, the first transformed luma reference **552***y*, the second transformed luma reference **554***y*, the first transformed chroma reference **542**, and the second transformed chroma reference **544**) can be concatenated (at step **595**) together to obtain input samples **580** for training the DNN and/or for prediction using the DNN. Beside the luma and chroma component, side information can be added to the input for training the neural network and/or prediction. For example, a QP value and block partition depth information can be used to generate a feature map that has the size N×M, and can be concatenated together (at step **595**) with image-based feature maps (e.g., the downsampled luma component **530***y*, the first transformed luma reference **552***y*, the second transformed luma reference **554***y*, the first transformed chroma reference **542**, and the second transformed chroma reference **544**) to generate the input samples **580** for training and/or prediction.
(60) A workflow of a process **600** in a general hybrid video coding system is described below with reference to FIG. **6**.
(61) A set of a reconstructed luma block **610** (also referred to as a luma component), side information **612**, adjacent luma references **614** to the luma block **610**, and adjacent chroma references **616** to the chroma block to be reconstructed, may be used as the input of a DNN **620**, so that a model of embodiments of the present disclosure can perform both training and predicting. The output **630** of the DNN **620** may be a predicted chroma component, and two chroma channels may be predicted using different DNN models or the same DNN model.
(62) According to embodiments, the inputs to the DNN **620** may be the input samples **580** described with reference to FIG. **5**. For example, the reconstructed luma block **610** may be the downsampled luma component **530***y*, the adjacent luma references **614** may be one or more of the first transformed luma reference **552***y* and the second transformed luma reference **554***y*, and the adjacent chroma references **616** may be one or more of first transformed chroma reference **542** and the second transformed chroma reference **544** (for one or both of the chroma channels **520***u* and **520***t*). According to embodiments, the side information may include, for example, the QP value and block partition depth information.
(63) The combination, the concatenation, or the order of how the reconstructed luma block **610**, the side information **612**, the adjacent luma references **614**, and the adjacent chroma references **616** are used as the input can be changed variously. According to embodiments, the side information **612**, the adjacent luma references **614**, and/or the adjacent chroma references **616** may be optional inputs for the DNN **620**, based on a decision(s) by the coding systems of embodiments of the present disclosure.
(64) According to embodiments, coding systems of the present disclosure may compute reconstruction quality (step **640**) by, for example, comparing the output **630** of the DNN **620** (e.g.,

the predicted chroma component) with the original chroma block **660**, and by comparing one or more chroma blocks from other predication modes (step **650**) with the original chroma block **660**. Based on determining one from among the output **630** (e.g., the predicted chroma component) and the one or more chroma blocks from the other predication modes (step **650**) has a highest reconstruction quality (e.g., closest to the original chroma block **660**), such block (or mode) may be selected by the coding system to be the reconstructed chroma block **670**.

(65) According to embodiments, at least one processor and memory storing computer program instructions may be provided. The computer program instructions, when executed by the at least one processor, may implement a system that performs any number of the functions described in the present disclosure. For example, with reference to FIG. **7**, the at least one processor may implement a system **700**. The system **700** may include a DNN(s) and at least one model thereof. The computer program instructions may include, for example, DNN code **710**, input generating code **720**, inputting code **730**, predicting code **740**, reconstruction quality code **750**, and image obtaining code **760**.

(66) The DNN code **710** may be configured to cause the at least one processor to implement the DNN(s) (and models thereof), according to embodiments of the present disclosure.

(67) The input generating code **720** may be configured to cause the at least one processor to generate inputs for the DNN(s), according to embodiments of the present disclosure (e.g., refer to descriptions of FIG. **5**). For example, the input generating code **720** may cause the processes described with reference to FIG. **5** to be performed.

(68) The inputting code **730** may be configured to cause the at least one processor to input the inputs into the DNN(s), according to embodiments of the present disclosure (e.g., refer to descriptions of the inputs into DNN **620** illustrated in FIG. **6**). For example, with reference to FIG. **6**, the inputs may include the reconstructed luma block **610**, the side information **612**, the luma references **614**, and/or the chroma references **616**.

(69) The predicting code **740** may be configured to cause the at least one processor to predict, by the DNN(s), a reconstructed chroma block, according to embodiments of the present disclosure (e.g., refer to descriptions of output **630** illustrated in FIG. **6**).

(70) The reconstruction quality code **750** may be configured to cause the at least one processor to compute reconstruction qualities of the reconstructed chroma block that is predicted by the DNN, and of another reconstructed chroma block(s) that is predicted using a different predication mode(s), according to embodiments of the present disclosure (e.g., refer to descriptions of steps **640** and **650** illustrated in FIG. **6**).

(71) The image obtaining code **760** may be configured to cause the at least one processor to obtain the image using the reconstructed chroma block that is predicted by the DNN, or the another reconstructed chroma block(s) that is predicted using a different predication mode(s), in accordance with embodiments of the present disclosure (e.g., refer to descriptions of step **640** and the reconstructed chroma block **670** illustrated in FIG. **6**). For example, the image obtaining code **760** may be configured to cause the at least one processor to select one from among the reconstructed chroma block and the another reconstructed chroma block, based on the one with the highest computed reconstruction quality, and use such reconstructed chroma block to obtain the image. According to embodiments, the image obtaining code **760** may be configured to cause the at least one processor to obtain the image using the reconstructed chroma block that is predicted by the DNN, without reconstruction qualities being computed and/or being used to select between reconstructed chroma blocks. According to embodiments, the reconstructed luma block may also be used to obtain the image.

(72) In comparison with prior cross component prediction methods in intra prediction mode, embodiments of the present disclosure provide a variety of benefits. For example, embodiments of the present disclosure provide a flexible and general framework that accommodates various shapes of reconstructed blocks. Also, embodiments of the present disclosure include an aspect of

exploiting the transformation mechanisms with various input information, thereby optimizing the learning ability of DNN models so as to improve coding efficiency. Furthermore, side information may be used with DNN to improve prediction results.

(73) The techniques of embodiments of the present disclosure described above, can be implemented as computer software using computer-readable instructions and physically stored in one or more computer-readable media. For example, FIG. **8** shows a computer system **900** suitable for implementing embodiments of the disclosed subject matter.

(74) The computer software can be coded using any suitable machine code or computer language, that may be subject to assembly, compilation, linking, or like mechanisms to create code comprising instructions that can be executed directly, or through interpretation, micro-code execution, and the like, by computer central processing units (CPUs), Graphics Processing Units (GPUs), and the like.

(75) The instructions can be executed on various types of computers or components thereof, including, for example, personal computers, tablet computers, servers, smartphones, gaming devices, internet of things devices, and the like.

(76) The components shown in FIG. **8** for computer system **900** are exemplary in nature and are not intended to suggest any limitation as to the scope of use or functionality of the computer software implementing embodiments of the present disclosure. Neither should the configuration of components be interpreted as having any dependency or requirement relating to any one or combination of components illustrated in the exemplary embodiment of a computer system **900**.

(77) Computer system **900** may include certain human interface input devices. Such a human interface input device may be responsive to input by one or more human users through, for example, tactile input (such as: keystrokes, swipes, data glove movements), audio input (such as: voice, clapping), visual input (such as: gestures), olfactory input (not depicted). The human interface devices can also be used to capture certain media not necessarily directly related to conscious input by a human, such as audio (such as: speech, music, ambient sound), images (such as: scanned images, photographic images obtain from a still image camera), video (such as two-dimensional video, three-dimensional video including stereoscopic video).

(78) Input human interface devices may include one or more of (only one of each depicted): keyboard **901**, mouse **902**, trackpad **903**, touch screen **910**, data-glove, joystick **905**, microphone **906**, scanner **907**, and camera **908**.

(79) Computer system **900** may also include certain human interface output devices. Such human interface output devices may be stimulating the senses of one or more human users through, for example, tactile output, sound, light, and smell/taste. Such human interface output devices may include tactile output devices (for example tactile feedback by the touch-screen **910**, data-glove, or joystick **905**, but there can also be tactile feedback devices that do not serve as input devices). For example, such devices may be audio output devices (such as: speakers **909**, headphones (not depicted)), visual output devices (such as screens **910** to include CRT screens, LCD screens, plasma screens, OLED screens, each with or without touch-screen input capability, each with or without tactile feedback capability—some of which may be capable to output two dimensional visual output or more than three dimensional output through means such as stereographic output; virtual-reality glasses (not depicted), holographic displays and smoke tanks (not depicted)), and printers (not depicted).

(80) Computer system **900** can also include human accessible storage devices and their associated media such as optical media including CD/DVD ROM/RW **920** with CD/DVD or the like media **921**, thumb-drive **922**, removable hard drive or solid state drive **923**, legacy magnetic media such as tape and floppy disc (not depicted), specialized ROM/ASIC/PLD based devices such as security dongles (not depicted), and the like.

(81) Those skilled in the art should also understand that term “computer readable media” as used in connection with the presently disclosed subject matter does not encompass transmission media,

carrier waves, or other transitory signals.
(82) Computer system **900** can also include interface to one or more communication networks. Networks can for example be wireless, wireline, optical. Networks can further be local, wide-area, metropolitan, vehicular and industrial, real-time, delay-tolerant, and so on. Examples of networks include local area networks such as Ethernet, wireless LANs, cellular networks to include GSM, 3G, 4G, 5G, LTE and the like, TV wireline or wireless wide area digital networks to include cable TV, satellite TV, and terrestrial broadcast TV, vehicular and industrial to include CANBus, and so forth. Certain networks commonly require external network interface adapters that attached to certain general purpose data ports or peripheral buses **949** (such as, for example USB ports of the computer system **900**; others are commonly integrated into the core of the computer system **900** by attachment to a system bus as described below (for example Ethernet interface into a PC computer system or cellular network interface into a smartphone computer system). Using any of these networks, computer system **900** can communicate with other entities. Such communication can be uni-directional, receive only (for example, broadcast TV), uni-directional send-only (for example CANbus to certain CANbus devices), or bi-directional, for example to other computer systems using local or wide area digital networks. Such communication can include communication to a cloud computing environment **955**. Certain protocols and protocol stacks can be used on each of those networks and network interfaces as described above.
(83) Aforementioned human interface devices, human-accessible storage devices, and network interfaces **954** can be attached to a core **940** of the computer system **900**.
(84) The core **940** can include one or more Central Processing Units (CPU) **941**, Graphics Processing Units (GPU) **942**, specialized programmable processing units in the form of Field Programmable Gate Areas (FPGA) **943**, hardware accelerators **944** for certain tasks, and so forth. These devices, along with Read-only memory (ROM) **945**, Random-access memory **946**, internal mass storage such as internal non-user accessible hard drives, SSDs, and the like **947**, may be connected through a system bus **948**. In some computer systems, the system bus **948** can be accessible in the form of one or more physical plugs to enable extensions by additional CPUs, GPU, and the like. The peripheral devices can be attached either directly to the core's system bus **948**, or through a peripheral bus **949**. Architectures for a peripheral bus include PCI, USB, and the like. A graphics adapter **950** may be included in the core **940**.
(85) CPUs **941**, GPUs **942**, FPGAs **943**, and accelerators **944** can execute certain instructions that, in combination, can make up the aforementioned computer code. That computer code can be stored in ROM **945** or RAM **946**. Transitional data can be also be stored in RAM **946**, whereas permanent data can be stored for example, in the internal mass storage **947**. Fast storage and retrieve to any of the memory devices can be enabled through the use of cache memory, that can be closely associated with one or more CPU **941**, GPU **942**, mass storage **947**, ROM **945**, RAM **946**, and the like.
(86) The computer readable media can have computer code thereon for performing various computer-implemented operations. The media and computer code can be those specially designed and constructed for the purposes of the present disclosure, or they can be of the kind well known and available to those having skill in the computer software arts.
(87) As an example and not by way of limitation, the computer system **900** having architecture, and specifically the core **940** can provide functionality as a result of processor(s) (including CPUs, GPUs, FPGA, accelerators, and the like) executing software embodied in one or more tangible, computer-readable media. Such computer-readable media can be media associated with user-accessible mass storage as introduced above, as well as certain storage of the core **940** that are of non-transitory nature, such as core-internal mass storage **947** or ROM **945**. The software implementing various embodiments of the present disclosure can be stored in such devices and executed by core **940**. A computer-readable medium can include one or more memory devices or chips, according to particular needs. The software can cause the core **940** and specifically the

processors therein (including CPU, GPU, FPGA, and the like) to execute particular processes or particular parts of particular processes described herein, including defining data structures stored in RAM **946** and modifying such data structures according to the processes defined by the software. In addition or as an alternative, the computer system can provide functionality as a result of logic hardwired or otherwise embodied in a circuit (for example: accelerator **944**), which can operate in place of or together with software to execute particular processes or particular parts of particular processes described herein. Reference to software can encompass logic, and vice versa, where appropriate. Reference to a computer-readable media can encompass a circuit (such as an integrated circuit (IC)) storing software for execution, a circuit embodying logic for execution, or both, where appropriate. The present disclosure encompasses any suitable combination of hardware and software.
(88) While this disclosure has described several non-limiting example embodiments, there are alterations, permutations, and various substitute equivalents, which fall within the scope of the disclosure. It will thus be appreciated that those skilled in the art will be able to devise numerous systems and methods which, although not explicitly shown or described herein, embody the principles of the disclosure and are thus within the spirit and scope thereof.

Claims

1. A method performed by at least one processor, the method comprising: obtaining reference components and side information associated with a reconstructed luma block, the side information includes block partition depth information; and predicting, by a deep neural network (DNN), a reconstructed chroma block based on the reconstructed luma block, the reference components, and the side information.
2. The method of claim 1, further comprising: obtaining the reconstructed luma block, wherein the reconstructed luma block is of an image or video; inputting the reconstructed luma block into a deep neural network (DNN); and inputting the reference components and the side information into the DNN.
3. The method of claim 1, wherein the reference components comprise at least one from among an adjacent luma reference of the reconstructed luma block and an adjacent chroma reference of the reconstructed chroma block, and the predicting further comprises predicting, by the DNN, the reconstructed chroma block based on the reconstructed luma block and the least one from among the adjacent luma reference and the adjacent chroma reference that are input.
4. The method of claim 1, wherein the reference components comprise an adjacent luma reference of the reconstructed luma block and an adjacent chroma reference of the reconstructed chroma block, and the predicting further comprises predicting, by the DNN, the reconstructed chroma block based on the reconstructed luma block, the adjacent luma reference, and the adjacent chroma reference.
5. The method of claim 1, further comprising generating a feature map based on the side information and concatenating the generated feature map with other image-based feature maps for DNN training.
6. The method of claim 1, wherein the side information includes at least one from among a quantization parameter (QP) value.
7. The method of claim 1, further comprising: computing reconstruction qualities of the reconstructed chroma block that is predicted by the DNN, and of another reconstructed chroma block that is predicted using a different predication mode; and obtaining an image or video using one from among the reconstructed chroma block that is predicted by the DNN and the another reconstructed chroma block that is predicted using the different predication mode, based on a highest one from among the reconstruction qualities computed.
8. The method of claim 1, further comprising: generating inputs of the DNN, wherein the

predicting comprises predicting, by the DNN, the reconstructed chroma block based on the inputs, wherein the generating comprises: reconstructing a luma block and obtaining adjacent luma references of the luma block; downsampling the luma block to obtain the reconstructed luma block as one of the inputs; downsampling the adjacent luma references of the luma block; and transforming the adjacent luma references, that are downsampled, to have a same size as the luma block that is downsampled, and wherein the inputs of the DNN include the luma block that is downsampled and the adjacent luma references that are transformed.

9. The method of claim 8, wherein the luma block is a 2N by 2M block, and the adjacent luma references comprise a 2N by 2K first luma reference block and a 2K by 2M second luma reference block for reference, wherein N, K, and M are integers, 2N is a width, 2M is a height, 2K is a number of rows or columns in luma reference, wherein the reconstructed luma block obtained by downsampling the luma block has a size of N×M, and wherein after downsampling the adjacent luma references, the first luma reference block has a size of N by K, and the second reference luma block has a size of K by M.

10. The method of claim 8, wherein the generating further comprises transforming adjacent chroma references of the reconstructed chroma block to be predicted to have the same size as the luma block that is downsampled, wherein the inputs of the DNN include the luma block that is downsampled, the adjacent luma references that are transformed, and the adjacent chroma references that are transformed.

11. A system comprising: at least one memory configured to store computer program code; and at least one processor configured to access the computer program code and operate as instructed by the computer program code, the computer program code comprising: obtaining code configured to cause the at least one processor to obtain reference components and side information associated with a reconstructed luma block, the side information includes block partition depth information; and predicting code configured to cause the at least one processor to predict, by a deep neural network (DNN) that is implemented by the at least one processor, a reconstructed chroma block based on the reconstructed luma block, the reference components, and the side information that are input.

12. The system of claim 11, wherein the reference components comprise at least one from among an adjacent luma reference of the reconstructed luma block and an adjacent chroma reference of the reconstructed chroma block, and the predicting code is further configured to cause the at least one processor to predict, by the DNN, the reconstructed chroma block based on the reconstructed luma block and the least one from among the adjacent luma reference and the adjacent chroma reference.

13. The system of claim 11, wherein the reference components comprise an adjacent luma reference of the reconstructed luma block and an adjacent chroma reference of the reconstructed chroma block, and the predicting code is further configured to cause the at least one processor to predict, by the DNN, the reconstructed chroma block based on the reconstructed luma block, the adjacent luma reference, and the adjacent chroma reference.

14. The system of claim 11, wherein the computer program code further comprises input generating code that is configured to cause the at least one processor to generate a feature map based on the side information and concatenate the generated feature map with other image-based feature maps for DNN training.

15. The system of claim 11, wherein the side information comprises a quantization parameter (QP) value.

16. The system of claim 11, wherein the computer program code further comprises: reconstruction quality code configured to cause the at least one processor to compute reconstruction qualities of the reconstructed chroma block that is predicted by the DNN, and of another reconstructed chroma block that is predicted using a different predication mode; and image obtaining code configured to cause the at least one processor to obtain by using one from among the reconstructed chroma block that is predicted by the DNN and the another reconstructed chroma block that is predicted using the

different predication mode, based on a highest one from among the reconstruction qualities computed.

17. The system of claim 11, wherein the computer program code further comprises: input generating code configured to cause the at least one processor to generate inputs of the DNN, wherein the predicting code is further configured to cause the at least one processor to predict, by the DNN, the reconstructed chroma block based on the inputs, wherein the input generating code is configured to cause the at least one processor to generate the inputs of the DNN by causing the at least one processor to: reconstruct a luma block and obtaining adjacent luma references of the luma block; downsample the luma block to obtain the reconstructed luma block as one of the inputs; downsample the adjacent luma references of the luma block; and transform the adjacent luma references, that are downsampled, to have a same size as the luma block that is downsampled, and wherein the inputs of the DNN comprise the luma block that is downsampled and the adjacent luma references that are transformed.

18. The system of claim 17, wherein the luma block is a 2N by 2M block, and the adjacent luma references comprise a 2N by 2K first luma reference block and a 2K by 2M second luma reference block for reference, wherein N, K, and M are integers, 2N is a width, 2M is a height, 2K is a number of rows or columns in luma reference.

19. The system of claim 18, wherein the reconstructed luma block obtained by downsampling the luma block has a size of $N \times M$, and after downsampling the adjacent luma references, the first luma reference block has a size of N by K, and the second reference luma block has a size of K by M.

20. A non-transitory computer-readable medium storing computer code that is configured to, when executed by at least one processor, cause the at least one processor to: input a reconstructed luma block, obtain reference components and side information associated with a reconstructed luma block, the side information includes block partition depth information; and predict, by a deep neural network (DNN), a reconstructed chroma block based on the reconstructed luma block, the reference components, and the side information.